

GIWON SONG

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Seoul, South Korea

EDUCATION

Seoul National University

Bachelor of Science in Electronic and Computer Engineering | 2029 (expected)

Core courses

Programming Methodology

Digital Logic Design and Lab.

Introduction to Quantum Computing (CSE)

Korea Science Academy of KAIST (KSA)

High School Diploma | 2023

MILITARY SERVICE

Republic of Korea Army (ROKA)

Technical Specialist & Cryptanalyst | May 2025 – Nov 2026

- > Decrypted complex ciphers by training a large language model (LLM) targeted to the specific properties of the plaintext.
- > Designed and delivered technical seminars on quantum computing fundamentals and quantum algorithms to 30+ military personnel, bridging advanced theoretical physics with accessible education.

RESEARCH EXPERIENCE

Quantum Information and Quantum Computer Lab (QuIQCL), Seoul National University

Undergraduate Research Intern | July 2024 – Apr 2025

- > Conducted research on the theoretical limits of quantum computing and algorithmic efficiency.
- > Focused on the classical simulation of quantum circuits utilizing the Extended Stabilizer Simulation framework.
- > Collaborated with a senior researcher to design and optimize computational models for state vector mathematics.

Undergraduate Research Program (URP) Participant | July 2023 – Aug 2023

- > Co-developed IQUIP (Ion trap Quantum computing User Interface Project), an interface and control software system for a laser projection setup within a vacuum chamber, facilitating precise hardware manipulation of an ion-trap quantum computer.
- > Project Repository: <https://github.com/snu-quiqcl/iquip>

Korea Science Academy of KAIST (KSA)

Student Researcher | 2021

- > Researched advanced implementations of Neural Radiance Fields (NeRF) for 3D scene representation and novel view synthesis.
- > Co-authored a formal research paper detailing the architectural improvements to the baseline NeRF model.

PUBLICATIONS & PRESENTATIONS

Manuscript in Preparation

Hwang, Y.^{*}, Song, G.^{*}, Park, K., Kwon, H.[†], Kim, T.[†] “Efficient expectation value estimation for quantum circuits via extended stabilizer frameworks and adaptive variance estimation” (Manuscript in preparation).

Publications

Park, K., Choi, K., Kim, M., Song, G., Kim, T.[†] “Quantum Linear Multistep Method for Using a Quantum Oracle with Differential Equations.” ArXiv preprint, 2025. ArXiv:<https://arxiv.org/abs/2501.03781>

Kim, B.* , Song, G.* , Cho, Y.* , Min, D., Kim, M., Hwang, S.† “MultiNeRF: Neural Radiance Field for Multiple Objects.” KSAREd, 2021.

Project Page: <https://giwon2004.github.io/multinerf/>

**These authors contributed equally to the work. †Corresponding author(s).*

Conference Presentations

Presenter, "Development of a Space-Efficient Quantum Circuit Simulator Optimized for Expectation Value Calculation." 2025 Quantum Information Society of Korea (QISK) Annual Conference, Busan, South Korea.

AWARDS & HONORS

Academic Excellence Scholarship (Merit-based) | Seoul National University | 2023 – 2024

Special Prize (IBM Quantum) | Korea Q Hackathon | 2024

Advanced Badge (Completed all exercises) | IBM Quantum Spring Challenge | 2024

credly.com/badges/b2f7edca-e091-4fda-8cf8-9ea219084ba4/public-url

Director's Award, Quantum Information Research Support Center | Korea Q Hackathon | 2023

Silver Prize | Korea Olympiad in Informatics (KOI) | 2016

TECHNICAL SKILLS

Programming & Tools

C++, Python, Qiskit, LaTeX, Git

Research Areas

Quantum Algorithms, Classical Simulation, Extended Stabilizer Framework, Computational Complexity

LANGUAGES

Korean: Native

English: Professional Working Proficiency

iBT TOEFL: 5.5/6 (110/120)

Japanese: Professional Working Proficiency

JLPT N1: Results Expected August 2026

JLPT N2: 165/180